

# ANDREW NGUYEN

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## PROFESSIONAL EXPERIENCE

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### Coinbase

*Senior Software Engineer - Financial Engineering*

2026 - Present

- DRI (Directly Responsible Individual) for Balance Reconciliation, the data pipeline that tracks and reconciles balances across hundreds of billions of dollars held in Coinbase's bank accounts — partnering with Operations, Finance, Accounting, and Legal stakeholders to scope use cases and replace a process an 8-person team previously ran by hand as their full-time job.
- Automated transaction and balance-break detection that flags discrepancies in real time, cutting detection time from days to hours and alerting Operations teams to \$1M+ in missing or unreconciled funds each month.
- Built an agent harness that reasons across cross-system data and workflow context to pinpoint root cause amid a multitude of race conditions, surfaced to Operations via Slack for self-service remediation — cutting engineering on-call time supporting Ops from half a workweek to under a day, and diagnosis-to-remediation time from an hour to minutes.

### Amazon

*Software Development Engineer - Alexa AI Developer Tech*

2025 - 2026

- Engineer on the Alexa AI Developer Tech team — building partner-facing agents that navigated partner websites to manage full reservation lifecycles, and the internal platforms and SDKs that powered Alexa teams across the organization.
- Designed the core agentic system powering Alexa's web automation — an LLM-driven agent that browses partner websites, reasons over page state, and executes multi-step workflows to create bookings, modify reservations, and retrieve service status across structurally diverse partner sites.
- Reduced partner onboarding time 68% (38 → 12 weeks) by enabling partners to define integrations directly in natural language — reducing dependency on Alexa engineering and scaling annual partner integrations from 2 to 15.
- Eliminated manual prompt engineering for VLM-based agents by building an eval pipeline that tracked milestone and goal success rates, auto-generated prompt improvements, and opened PRs for review — saving ~2 weeks per partner integration and keeping agents accurate as partner websites evolved.
- Enabled 5 partner integrations by leading delivery of the Home Services SDK from design through launch across a 5-engineer team — owning milestone planning, task scoping, and coordination across product, design, and partner engineering.

### Amazon Web Services

*Software Development Engineer - AWS Wickr*

2022 - 2025

- Replaced Wickr's legacy portal with an AWS Console management experience built in React, giving 40+ enterprise and government customers self-service control over networks, users, configuration, and billing after AWS's acquisition — eliminating dependency on the pre-acquisition platform.
- Owned Wickr's AWS-native billing integration from design through migration — architected a metering microservice across individual payers and organizations, migrated existing customers from the legacy billing API, and eliminated third-party vendor fees.

### JPMorgan Chase

*Senior Associate Software Engineer*

2021 - 2022

### VARIOUS COMPANIES

*Lighting Automation Programmer*

2018 - 2021

## TECHNICAL SKILLS

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Languages: Typescript, Java, Python, Go, SQL

Technologies: Distributed Systems, AWS Services, AWS CDK, VLM Models, Agent Orchestration, NoSQL, Spark, Flink, Databricks, Snowflake

## OPEN SOURCE PROJECTS

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**ComfyUI AWS Queue Image/Video Generation, Distributed GPU Computation**

[github.com/AndrewNguyenn/comfyui-aws-queue](https://github.com/AndrewNguyenn/comfyui-aws-queue)

CDK-deployed, queue-driven ComfyUI pipeline on spot AWS GPU fleets — Lambda-dispatched jobs, per-workload ECS fleet specialization, and automated cost guardrails keeping spend near \$100/month.

## EDUCATION

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**Marymount Manhattan College**

New York City

BA Lighting Programming